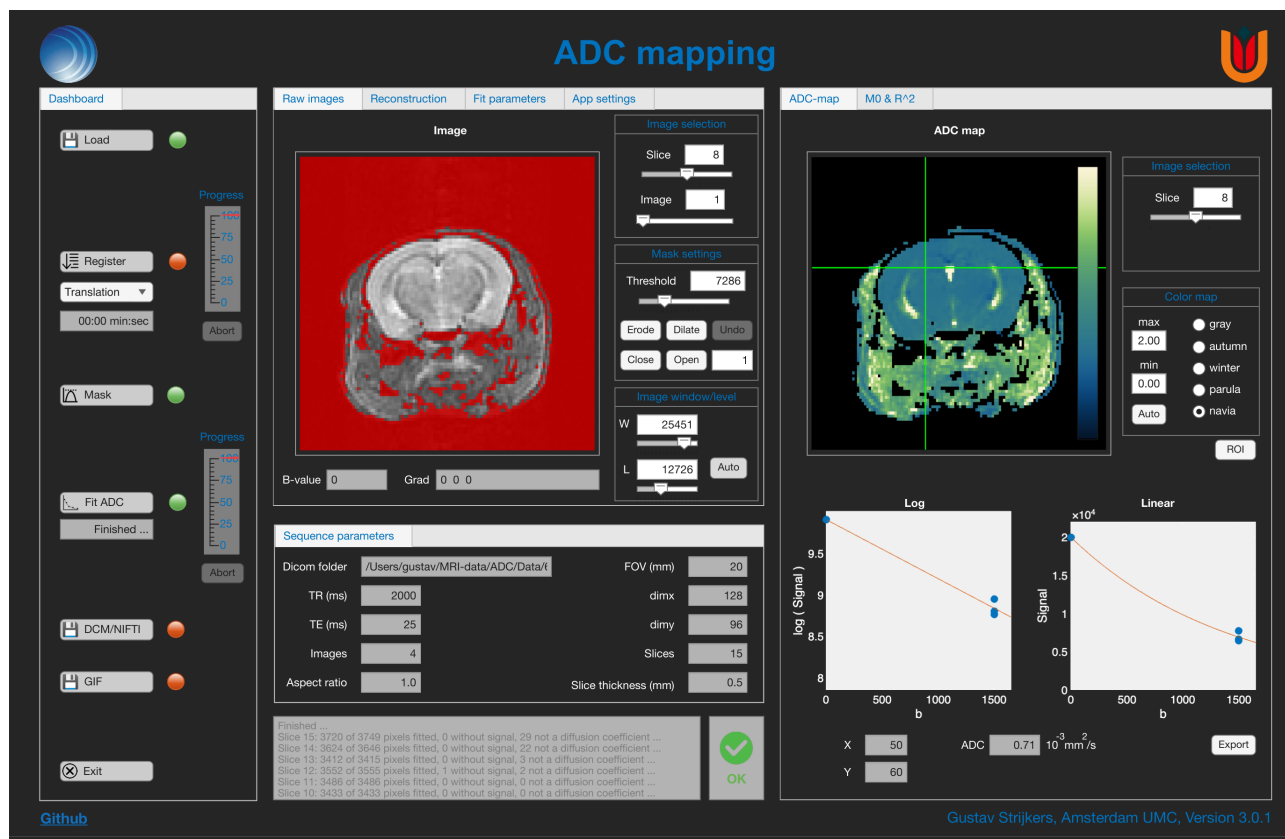




## Table of contents

1. Overview
2. Contact
3. Disclaimer
4. Installation notes
5. Running the software
6. Basic operation



## 1. Overview

ADCmapp is a software utility and graphical user interface to reconstruct apparent diffusion coefficient (ADC) maps from diffusion-weighted acquisitions with an MR Solutions preclinical MRI system. It reads a diffusion-weighted EPI scan from its MRD files, which it reconstructs, or a series of DICOM files.

Note: the program does not calculate a diffusion tensor, but only estimates the mean diffusivity / ADC.

## 2. Contact

ADCmapp and this manual are written by:

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## 3. Disclaimer

The software has been extensively tested using mouse and rat data acquired with an MR Solutions preclinical 7.0T/24cm system.

However, this does not warrant the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free.

In case of questions or issues, please contact Gustav Strijkers or Bram Coolen.



### 4. Installation notes

#### Software download

The Matlab app and a Windows standalone version (using the free Matlab runtime engine) can be downloaded from GitHub:

<https://github.com/Moby1971?tab=repositories>

#### Installation of the Windows standalone version

ADCmapp\_app\_installer.exe

Will install the Matlab runtime engine and the ADCmapp program.

#### Elastix registration

Image registration uses elastix, which comes with ADCmapp for macOS, Windows and Linux. The macOS version is built for Apple silicon; on an Intel Mac, which is no longer supported, it does not run. When elastix cannot be run, ADCmapp registers the images with Matlab's imregtform instead, and says so at start.



## 5. Running the software

### Running in Matlab 2025a

The ADCmapp software can be started from its root directory from the command line.

```
>> ADCmapp
```

ADCmapp starts in about 13 seconds. A DICOM series of 15 slices of 128 x 128 with 4 images needs about 2.5 GB of memory. A DTI scan of 130 volumes of 34 slices of 70 x 72 needs about 16 GB when it is reconstructed from its MRD files and 10 GB when its DICOM series is read; 7 GB of that is the parallel pool the NIfTI conversion starts.

Additional licenses may be required.

```
MATLAB  
Image Processing Toolbox  
Signal Processing Toolbox
```

### Running the Windows standalone

The Windows standalone version can be run from the start menu or the desktop icon.

### Settings and log files

ADCmapp keeps its settings in a file, adcmappSettings.json, in a folder of its own for every version of the app:

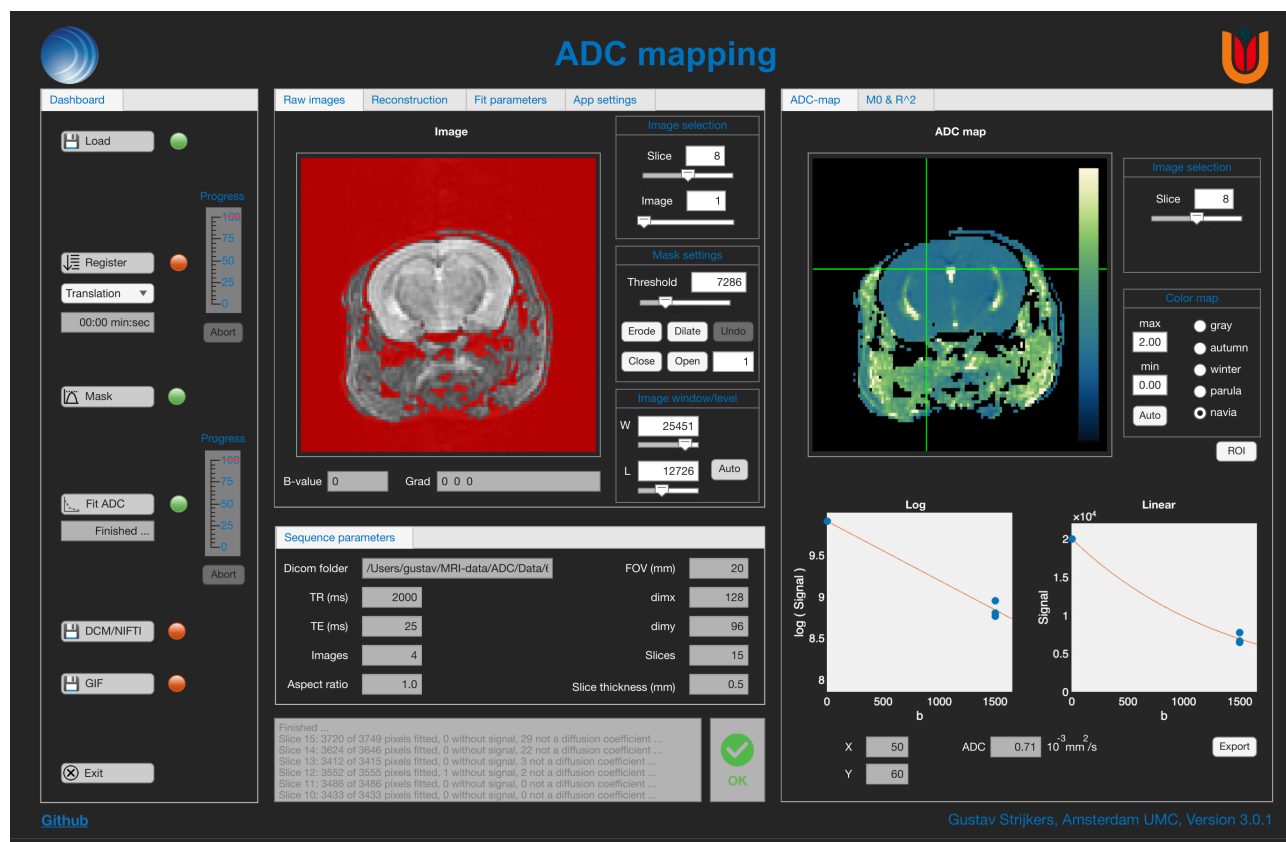
```
macOS      ~/Library/Application Support/ADCmapp/<version>  
Windows    %APPDATA%\ADCmapp\<version>  
Linux      ~/.config/ADCmapp/<version>
```

The file is written with the standard values the first time a version starts, and can then be edited by hand; changes take effect at the next start. It holds whether the app looks for a newer version, the folders where loading, the gradient table and the exports start, the color scheme, the sounds, the registration method, the corrections and the MP-PCA denoising of an MRD reconstruction (denoising of a DICOM series is off unless the file switches it on), the R-square cut-off and the color maps. A value that cannot be used is replaced by its standard value, and the message box says so. The color scheme and the sounds can also be switched in the App settings tab, for the session only.

Every session writes a log file, adcmapp-log-<date and time>.txt, to the same folder. The log keeps every message and the parameters of each registration and fit. Logs older than the newest 50 are removed at start.

## 6. Basic operation

The ADCmapp program operates from a single window with 5 panels.



### Panel 1: Dashboard

This panel contains the task buttons, with the progress gauges, times and Abort buttons of the registration, reconstruction and fit. The tasks run from top to bottom; registration and masking can be left out. A green light shows a task done, yellow a task running, and red one not done or failed.

### Panel 2: Raw images, Reconstruction, Fit parameters & App settings

This panel has 4 tabs: raw images, reconstruction, fit parameters and app settings.

### Panel 3: Sequence parameters

This panel displays the relevant acquisition parameters.

### Panel 4: ADC-map, M0 & R<sup>2</sup>

During and after fitting this panel shows the fitted ADC maps. M0 maps and R<sup>2</sup> goodness of fit maps are displayed in the second tab.

### Panel 5: Messages

Displays program messages and the status: OK, BUSY, WARNING or ERROR.

## Step 1: Loading data

Press  **Load** to load a scan: choose the folder of a diffusion-weighted EPI scan with its MRD files, or of a DICOM series. An MRD scan is first reconstructed, with the corrections ticked on the Reconstruction tab and MP-PCA denoising, and written as DICOM, NIfTI and BIDS; that series is then read (see MRD scans, after Step 6).

Relevant acquisition parameters will be shown in panel 3.

| Sequence parameters  |                                   |
|----------------------|-----------------------------------|
| Dicom folder         | /Users/gustav/MRI-data/ADC/Data/t |
| TR (ms)              | 2000                              |
| TE (ms)              | 25                                |
| Images               | 4                                 |
| Aspect ratio         | 1.0                               |
| FOV (mm)             | 20                                |
| dimx                 | 128                               |
| dimy                 | 96                                |
| Slices               | 15                                |
| Slice thickness (mm) | 0.5                               |

## Step 2: Image registration

Use the sliders and/or edit-fields to inspect the slices and images (b-values / diffusion-directions combinations). The window and level can be set with the W and L fields and sliders, or with Auto.

| Raw images                                                                                                                                                                                                                                                                            | Reconstruction | Fit parameters | App settings |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|--------------|
| <div>Image</div>  <div>           B-value 0    Grad 0 0 0         </div>                                                                                                                           |                |                |              |
| <div>Image selection</div> <div>           Slice 8<br/>  </div> <div>           Image 1<br/>  </div>            |                |                |              |
| <div>Mask settings</div> <div>           Threshold 0<br/>  </div> <div>           Erode Dilate Undo         </div> <div>           Close Open 1         </div>                                     |                |                |              |
| <div>Image window/level</div> <div>           W 25451<br/>  </div> <div>           L 12726    Auto<br/>  </div> |                |                |              |

If the images are not perfectly registered, press  **Register** to register every image to the first image with the lowest b-value, per slice, with elastix, as a translation, rigid, affine or B-spline transform, chosen in the list below the button.

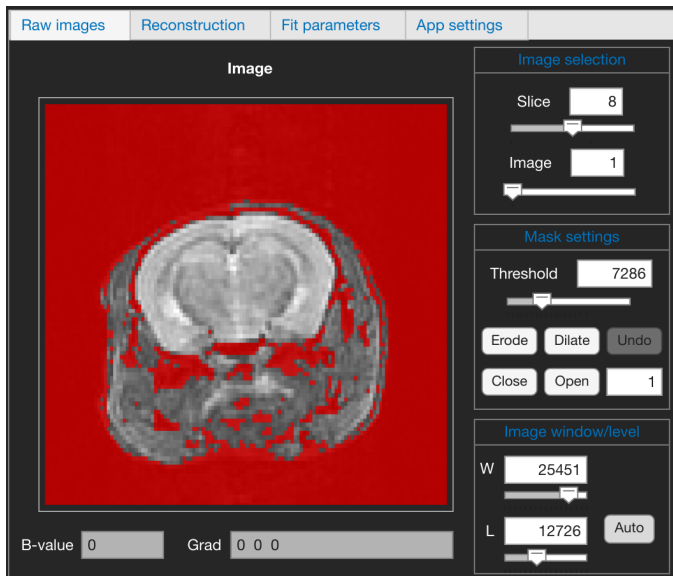
NOTE: this step usually can be omitted; Abort stops it.

### Step 3: Segmenting the images

To prevent unnecessary fitting of the background pixels, segment the images first; without a mask every pixel is fitted.

Press  Mask to mask them with an automatic (Otsu) threshold per slice.

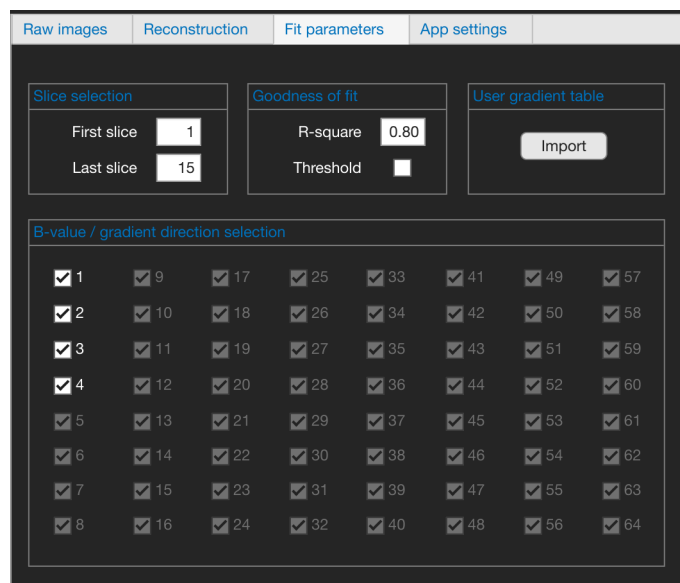
The result will look something like the figure below in which the red pixels indicate the regions that will be skipped during ADC fitting.



In case automatic thresholding is not optimal, the threshold can be adjusted per slice, with the threshold field or slider. The mask can be refined by erode, dilate, close and open operations on the red region, of the size in pixels in the field beside them; Undo takes the last one back. A registration afterwards clears the mask and the maps.

## Step 4: Fitting ADC maps

Press the Fit parameters tab to inspect the fit settings.

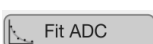


The first and last slice that will be fitted can be selected.

When the Threshold box is ticked, the ADC and M0 of a pixel whose  $R^2$  is below R-square are cleared; the  $R^2$  map keeps them, so the rejected pixels can be seen.

Note: If multiple gradient directions are used and tissue diffusivity is anisotropic  $R^2$  is not a good measure for the goodness of fit, because it will include variations due to the directional anisotropy.

The bottom part of this tab shows a list of all the images (b-values and diffusion-direction combinations). The images that are selected will be included in the fitting process. Import reads a gradient table: a text file with a line per image, the b-value and the three components of the direction, separated by spaces, tabs, commas or semicolons; % and # start comments.

Press  to start the ADC fitting.

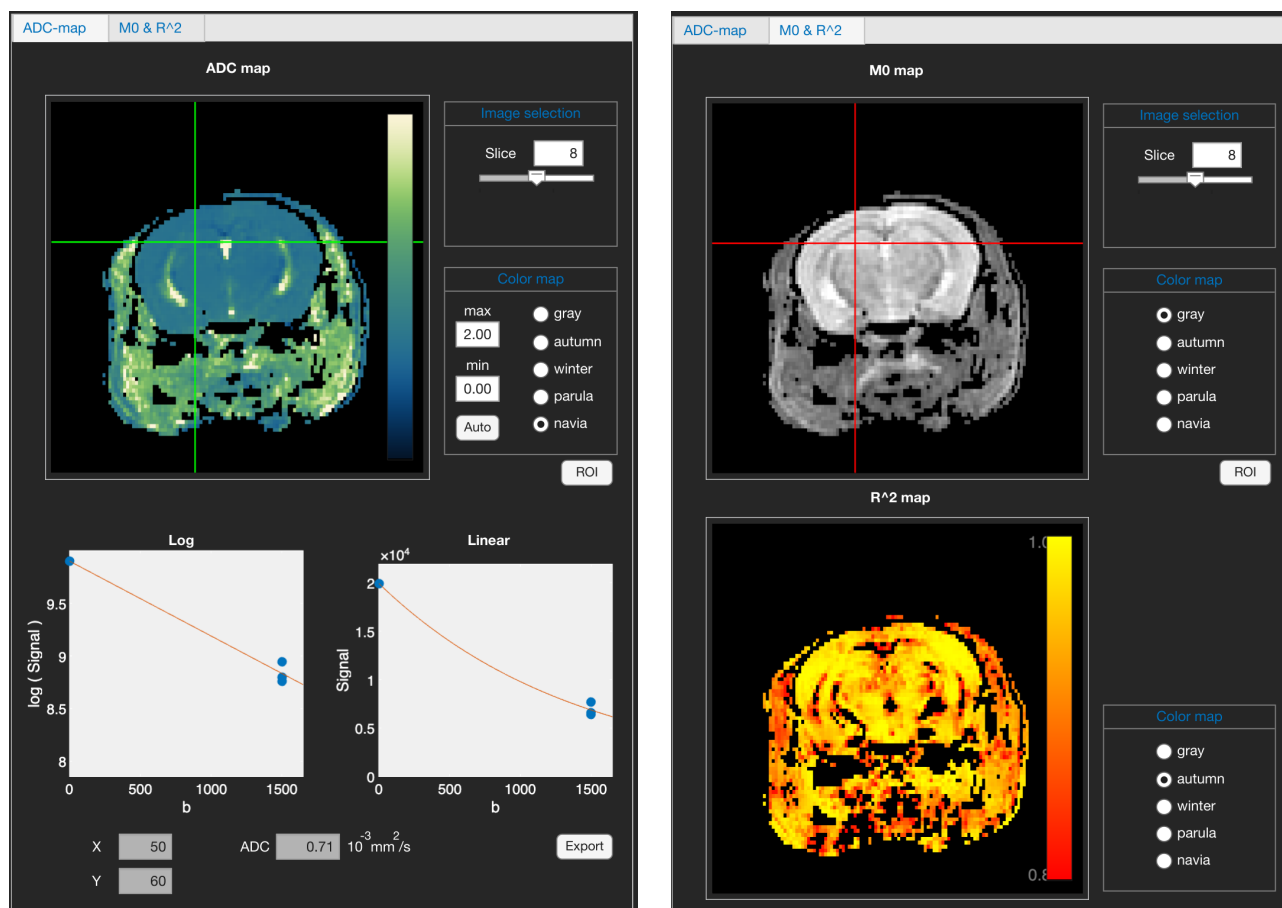
The selected b-values must take two values or more, at least 100 s/mm<sup>2</sup> apart; without a mask every pixel is fitted, with a warning. The fit is a least-squares line through  $\log(\text{Signal})$  against b, with the ADC in 10<sup>-3</sup> mm<sup>2</sup>/s:

$$\text{Signal} = M0 \exp(-b \cdot \text{ADC})$$



## Step 5: Fit results

The ADC maps will be shown in the ADC-map tab, on the scale set with max and min, or with Auto. An M0 map and R<sup>2</sup> map are shown in the second tab, each with its own color map.



Click on the ADC map with the mouse cursor to display  $\log(\text{Signal})$  and Signal as function of  $b$ -value (symbols) and the fit result (orange lines). The fitted ADC in the pixel is shown below the plots. A red symbol indicates that this  $b$ -value/diffusion-direction was omitted in the fitting process, according to the list in the Fit parameters tab (see step 4). With ROI, pixels are painted with the brush, of the size set beside it; the plots then show the mean signal of the ROI and the curve of its mean ADC and M0, and Erase clears it. Export writes the  $b$ -values and the signal under the cross-hair or the ROI to a CSV file.



## Step 6: Exporting the ADC maps

There are two ways to export the ADC, M0, and R<sup>2</sup> maps.

### (1) DCM/NIfTI



Exports the ADC, M0 and R<sup>2</sup> maps in DICOM and NIfTI format for further processing in 3rd party software, with a copy of the diffusion-weighted images.

### (2) GIF



Exports the three maps as animated GIF files, one frame per slice, with two pictures of the app's window.

The exports go beside the scanner's files, in .../Data/<study>/DICOM, Nifti and GIF, when the scan was loaded from the scanner's folders; otherwise the app asks for a folder. Every export gets a new numbered folder.

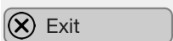
## MRD scans

The Load button takes the folder of a scan or of its study and prefers an MRD scan to a DICOM series; MRD scans must be 2D diffusion EPI with at least two b-values.

The Reconstruction tab holds the corrections: ramp sampling (off by default), POCS for partial Fourier, distortion correction with the study's reversed phase-encode scan, and registration of the volumes. They start as the settings file says and take effect at the next load; MP-PCA denoising is set in the settings file. Ghost and drift correction, which also takes out the frequency offset of a scan run off resonance, and a Tukey window against ringing are always applied, and masking when the study has a FLASH mask scan.

The reconstruction shows its progress on the registration gauge; Abort stops it, and nothing is loaded. The images are written into the study's Data folder, or one asked for: DICOM in DICOM/<series>P/<n> with recoparameters\_<tag>.txt, NIfTI in Nifti/<tag>P/<n> with .bval, .bvec and scripts for MRtrix3 and DSI Studio, and BIDS. Every load writes them again, into the next numbered folder.

## Step 7: Exit

Press  to shut down the program.